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SUBJECT : SE LAB-05

```
(venv) PS C:\Users\DELL\L5_45_ticket_booking_system> Get-Content .\bandit_report.txt
Run started:2026-10-14 06:22:06.134169+00:00

Test results:
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: .\.\ticket_booking_system.py:26:4
25              theater_seats[seat_id]["is_sold"] = False
26          except:
27              pass
28

--------------------------------------------------
>> Issue: [B307:blacklist] Use of possibly insecure function - consider using safer ast.literal_eval.
   Severity: Medium   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b307-eval
   Location: .\.\ticket_booking_system.py:42:18
41          base_price = theater_seats[seat_id]["price"]
42          final_price = eval(promo_expr)
43          return final_price

--------------------------------------------------
>> Issue: [B324:hashlib] Use of weak MD5 hash for security. Consider usedforsecurity=False
   Severity: High   Confidence: High
```

```
>> Issue: [B324:hashlib] Use of weak MD5 hash for security. Consider usedforsecurity=False
   Severity: High   Confidence: High
   CWE: CWE-327 (https://cwe.mitre.org/data/definitions/327.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b324_hashlib.html
   Location: .\.\ticket_booking_system.py:47:11
46      def generate_ticket_hash(ticket_code):
47          return hashlib.md5(ticket_code.encode()).hexdigest()
48

--------------------------------------------------
>> Issue: [B311:blacklist] Standard pseudo-random generators are not suitable for security/cryptographic purposes.
   Severity: Low   Confidence: High
   CWE: CWE-330 (https://cwe.mitre.org/data/definitions/330.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b311-random
   Location: .\.\ticket_booking_system.py:51:11
50      def generate_booking_pin():
51          return random.randint(1000, 9999)
52

--------------------------------------------------
>> Issue: [B605:start_process_with_a_shell] Starting a process with a shell, possible injection detected, security is
sue.
   Severity: High   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b605_start_process_with_a_shell.html
   Location: .\.\ticket_booking_system.py:58:4
57          f.close()
58          os.system("echo Seating chart exported to: " + filename)
```

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Code scanned:
        Total lines of code: 60
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 2
                Medium: 1
                High: 2
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 5
Files skipped (0):
```

Pylint and Flake8:-

```
(venv) PS C:\UserGet-Content .\pylint_report.txttem>
************* Module ticket_booking_system
ticket_booking_system.py:82:0: C0304: Final newline missing (missing-final-newline)
ticket_booking_system.py:15:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:15:0: C0103: Function name "reserveSeat" doesn't conform to snake_case naming style (invalid
-name)
ticket_booking_system.py:15:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
ticket_booking_system.py:19:25: C0209: Formatting a regular string which could be an f-string (consider-using-f-strin
g)
ticket_booking_system.py:23:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:26:4: W0702: No exception type(s) specified (bare-except)
ticket_booking_system.py:30:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:32:7: C0121: Comparison 'theater_seats.get(seat_id) == None' should be 'theater_seats.get(se
at_id) is None' (singleton-comparison)
ticket_booking_system.py:34:4: R1703: The if statement can be replaced with 'return bool(test)' (simplifiable-if-stat
ement)
ticket_booking_system.py:34:4: R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code ins
ide it (no-else-return)
ticket_booking_system.py:31:4: W0612: Unused variable 'booking_fee' (unused-variable)
ticket_booking_system.py:40:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:42:18: W0123: Use of eval (eval-used)
ticket_booking_system.py:41:4: W0612: Unused variable 'base_price' (unused-variable)
ticket_booking_system.py:46:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:50:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:54:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:55:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
ticket_booking_system.py:55:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
ticket_booking_system.py:61:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:61:24: W0621: Redefining name 'theater_seats' from outer scope (line 8) (redefined-outer-nam
e)
ticket_booking_system.py:63:8: W0612: Unused variable 'seat' (unused-variable)
```

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e)
ticket_booking_system.py:63:8: W0612: Unused variable 'seat' (unused-variable)
ticket_booking_system.py:69:0: C0116: Missing function or method docstring (missing-function-docstring)
ticket_booking_system.py:6:0: W0611: Unused import math (unused-import)

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Your code has been rated at 5.37/10

(venv) PS C:\Users\DELL\L5_45_ticket_booking_system> Get-Content .\flake8_report.txt
.\ticket_booking_system.py:6:1: F401 'math' imported but unused
.\ticket_booking_system.py:19:80: E501 line too long (82 > 79 characters)
.\ticket_booking_system.py:26:5: E722 do not use bare 'except'
.\ticket_booking_system.py:31:5: F841 local variable 'booking_fee' is assigned to but never used
.\ticket_booking_system.py:32:35: E711 comparison to None should be 'if cond is None:'
.\ticket_booking_system.py:41:5: F841 local variable 'base_price' is assigned to but never used
.\ticket_booking_system.py:82:7: W292 no newline at end of file
(venv) PS C:\Users\DELL\L5_45_ticket_booking_system>
```

1. Bandit final report:-

```
(venv) PS C:\Users\DELL\L5_45_ticket_booking_system> Get-Content .\bandit_report.txt
Run started:2026-10-14 06:39:30.263882+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 75
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
```

2. Pylint final report

```
(venv) PS C:\Users\DELL\L5_45_ticket_booking_system> Get-Content .\pylint_report.txt

 --------------------------------------------------------------------
 Your code has been rated at 10.00/10 (previous run: 9.83/10, +0.17)
```

3.Flake8 final report

```
(venv) PS C:\Users\DELL\L5_45_ticket_booking_system> python -m flake8 .\ticket_booking_system.py
(venv) PS C:\Users\DELL\L5_45_ticket_booking_system>
```

4.Program execution

```
(venv) PS C:\Users\DELL\L5_45_ticket_booking_system> python .\ticket_booking_system.py
Is B1 available: True
Promo price: 16.0
Ticket Hash: 3fab8fc159f6c8f4d993cfca213b143587ff0137b6bbd488716b060a1a33ab80
Ticket verification: True
Booking PIN: 1450
Seating chart exported to: seating_chart.txt
Sold count: 1
(venv) PS C:\Users\DELL\L5_45_ticket_booking_system>
```